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MEDOPENCLAW: Auditable Medical Imaging Agents Reasoning over Uncurated Full Studies

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Abstract

Currently, evaluating vision-language models (VLMs) in medical imaging tasks oversimplifies clinical reality by relying on pre-selected 2D images that demand significant manual labor to curate. This setup misses the core challenge of real-world diagnostics: a true clinical agent must actively navigate full 3D volumes across multiple sequences or modalities to gather evidence and ultimately support a final decision. To address this, we propose MEDOPENCLAW, an auditable runtime designed to let VLMs operate dynamically within standard medical tools or viewers (e.g., 3D Slicer). On top of this runtime, we introduce MEDFLOW-BENCH, a full-study medical imaging benchmark covering multi-sequence brain MRI and lung CT/PET. It systematically evaluates medical agentic capabilities across viewer-only, tool-use, and open-method tracks. Initial results reveal a critical insight: while state-of-the-art LLMs/VLMs (e.g., Gemini 3.1 Pro and GPT-5.4) can successfully navigate the viewer to solve basic study-level tasks, their performance paradoxically degrades when given access to professional support tools due to a lack of precise spatial grounding. By bridging the gap between static-image perception and interactive clinical workflows, MEDOPENCLAW and MEDFLOW-BENCH establish a reproducible foundation for developing auditable, full-study medical imaging agents.

1 Introduction

Medical vision-language models have progressed rapidly, from early medical VQA benchmarks [1, 2] to more recent medical multimodal systems and expert-level evaluation sets [3–5]. Yet much of their medical imaging evaluation still relies on a simplified proxy setting: the model is given one or a few **pre-selected diagnostically relevant 2D images** and asked to answer a localized question. This setup is useful for testing recognition on curated inputs, but it removes the central difficulty of medical imaging, especially radiology. It also keeps the decision process largely opaque: the model returns an answer, but not a replayable account of where it looked, what evidence it gathered, or how the final conclusion was reached.

This creates a substantial gap to real clinical workflow [6–8]. In practice, medical imaging analysis is a study-level process. A reader must inspect a full 3D examination, choose relevant series or modalities, navigate across many slices, adjust display settings such as windowing or fusion, compare evidence across views, and often perform measurements or specialized analysis before committing to an interpretation. Many clinically relevant findings are not visible in a single image. They emerge

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only across adjacent slices, across sequences, or after the relevant region has been localized [9–11]. A meaningful evaluation setting should therefore test not only whether an agent can produce the correct answer, but whether it can search a full study in a way that is transparent, replayable, and auditable.

To study this setting, we introduce two core components. First, we present **MEDOPENCLAW**, an auditable runtime that links between a backbone VLM model and the medical viewer platform, e.g. 3D Slicer [12] through a bounded set of viewer actions while keeping the execution trace visible. Second, we introduce **MEDFLOW-BENCH**, a benchmark for full-study medical imaging analysis episodes built on top of this runtime. Rather than evaluating models on isolated rendered images, MEDFLOW-BENCH evaluates whether they can inspect a full study, gather evidence through interaction, and produce answers under a controlled and reviewable study-level protocol. The current release covers multi-sequence brain MRI and lung CT/PET and is designed to support realistic, reproducible, and auditable evaluation of full-study medical reasoning.

Our results lead to a more important conclusion than simply showing that agents can interact with a viewer. Current VLM agents such as GPT-5.4 and Gemini-3.1-pro can already solve a meaningful portion of the study-level task by navigating the viewer directly, which suggests that viewer-native full-study reasoning is now feasible. However, the gap to real clinical workflow is still not closed. In particular, precise expert-tool use remains a bottleneck: access to professional analysis tools does not automatically translate into better performance, because current agents still lack the spatial grounding and control precision needed to use them reliably. In other words, moving from static-image benchmarks to real clinical-style study inspection is now possible, but robust end-to-end clinical workflow execution remains out of reach for current systems.

Our contributions are:

- We introduce **MEDOPENCLAW**, an auditable runtime that enables VLM agents to operate medical viewers, e.g. 3D Slicer, on full studies rather than answering questions on curated and pre-selected 2D inputs.
- We introduce **MEDFLOW-BENCH**, a more realistic and auditable benchmark for study-level medical imaging reasoning, designed to evaluate whether models can search, ground, and justify decisions over full clinical imaging studies.
- Using this benchmark, we show that the gap between current medical VLMs and real clinical workflow is **not yet closed**: viewer navigation is already feasible, but reliable quantitative tool use and clinically realistic end-to-end execution remain limited by weak spatial grounding and control.

2 Related Work

Static medical VQA and medical VLM benchmarks. Early medical VQA datasets established language-conditioned evaluation on medical images [1, 13, 14, 2, 15]. Subsequent medical QA and medical multimodal benchmarks broadened task scope, reasoning difficulty, and clinical coverage [3–5, 16–19]. Related lines such as medical report generation and image-grounded medical question answering also largely assume fixed image inputs rather than study-level interaction [20–23]. Collectively, these settings have been valuable for measuring visual recognition, medical knowledge use, and language-conditioned reasoning on curated inputs. However, they typically begin from one or a few pre-selected diagnostically relevant 2D views rather than from a full imaging study. Our work is motivated by this gap to study whether we can perform auditable study-level reasoning over full practical clinical imaging studies.

Medical agents. Another line of work studies medical agents that perform multi-step reasoning [24–27] and evidence gathering [28–30]. Some focus on radiology-oriented agents, especially for chest X-ray analysis or reporting [20, 31], while others study broader multimodal medical agents that select among specialized tools or APIs across tasks [32, 33]. These systems make the reasoning loop more explicit than standard static benchmarks, but many still operate on fixed images, isolated APIs, or abstracted tool interfaces rather than continuous interaction with a viewer over a full tomographic exam.

Full-study and interactive medical imaging systems. More closely related work begins to address study-level or interactive radiology settings. One line studies volumetric or 3D medical image reasoning with language-guided analysis over image volumes [34–36]. Another line evaluates agents

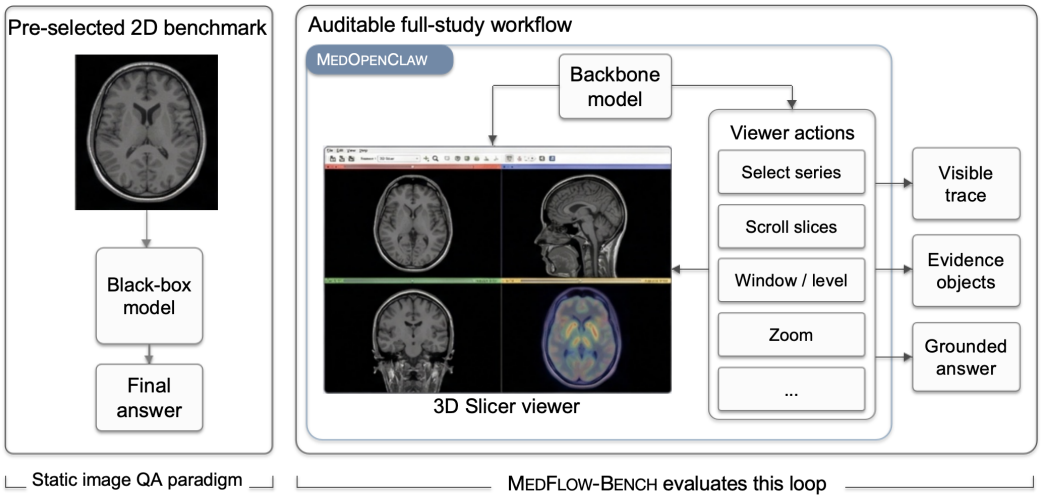

Figure 1: **Left**: Conventional medical VQA benchmarks rely on pre-selected, diagnostically relevant 2D images as inputs. They evaluate black-box models, where neither the decision-making process nor the supporting evidence is observable. **Right**: In contrast, our proposed benchmark is built on our runtime, MEDOPENCLAW, which interacts with 3D Slicer through a bounded viewer interface. This setup produces a transparent reasoning process, including an explicit trace, evidence objects, and a grounded final answer. MEDFLOW-BENCH is designed to evaluate this interactive loop. Moreover, toolkits such as MONAI [45] can be seamlessly integrated into both the framework and the benchmark. Beyond evaluation, the entire system can function as a medical imaging copilot, MEDCOPILOT, assisting clinicians and alleviating the complexity of real-world workflows.

in simulated or simplified radiology environments [37, 38]. A third line explores natural-language assistance or copilot-style interaction inside existing imaging software such as 3D Slicer [39, 40]. These directions move closer to clinical imaging workflow, but differ in whether the main emphasis is on volumetric reasoning, environment design, or software-integrated interaction. Our work sits at the intersection of these threads by focusing on study-level evaluation in a real viewer with preserved execution traces.

General-purpose agent runtimes. At the systems level, MEDOPENCLAW is related to general-purpose agent runtimes such as OpenClaw, which support heterogeneous tools and channels [41, 42] and target open-world task completion [43, 44]. In contrast to these runtimes, MEDOPENCLAW builds bounded control into the task interface itself: agents are restricted to viewer-native operations and vetted analysis tools under a standardized study-level protocol. This makes the system safer, more auditable, and better aligned with clinical workflow.

3 MEDOPENCLAW Runtime

MEDOPENCLAW is a runtime and API layer, not a model. It sits between a backbone VLM agent and medical tools or viewers, e.g., 3D Slicer [12], runs externally without modifying the viewer source code, and exposes a fixed interface for study inspection. Through this interface, the agent can perform the same core operations as a human reader inside the viewer, including selecting series, scrolling through slices, adjusting window or fusion settings, bookmarking views, taking measurements, and exporting evidence. Figure 1 summarizes this contract.

To structure this interaction, the exposed action space is organized into three distinct layers. First, primitive viewer actions support essential navigation and display control, such as selecting series and scrolling slices. Second, evidence operations allow the agent to capture and export reviewable artifacts, including bookmarked views, drawn masks, and measurement logs. Third, optional expert tools facilitate advanced segmentation or quantitative analysis. For instance, the MONAI-based reference tool pack [45] operates exclusively at this third layer. Crucially, this tiered architecture is not just a software abstraction; it directly informs our evaluation design. As detailed later in Section 4,

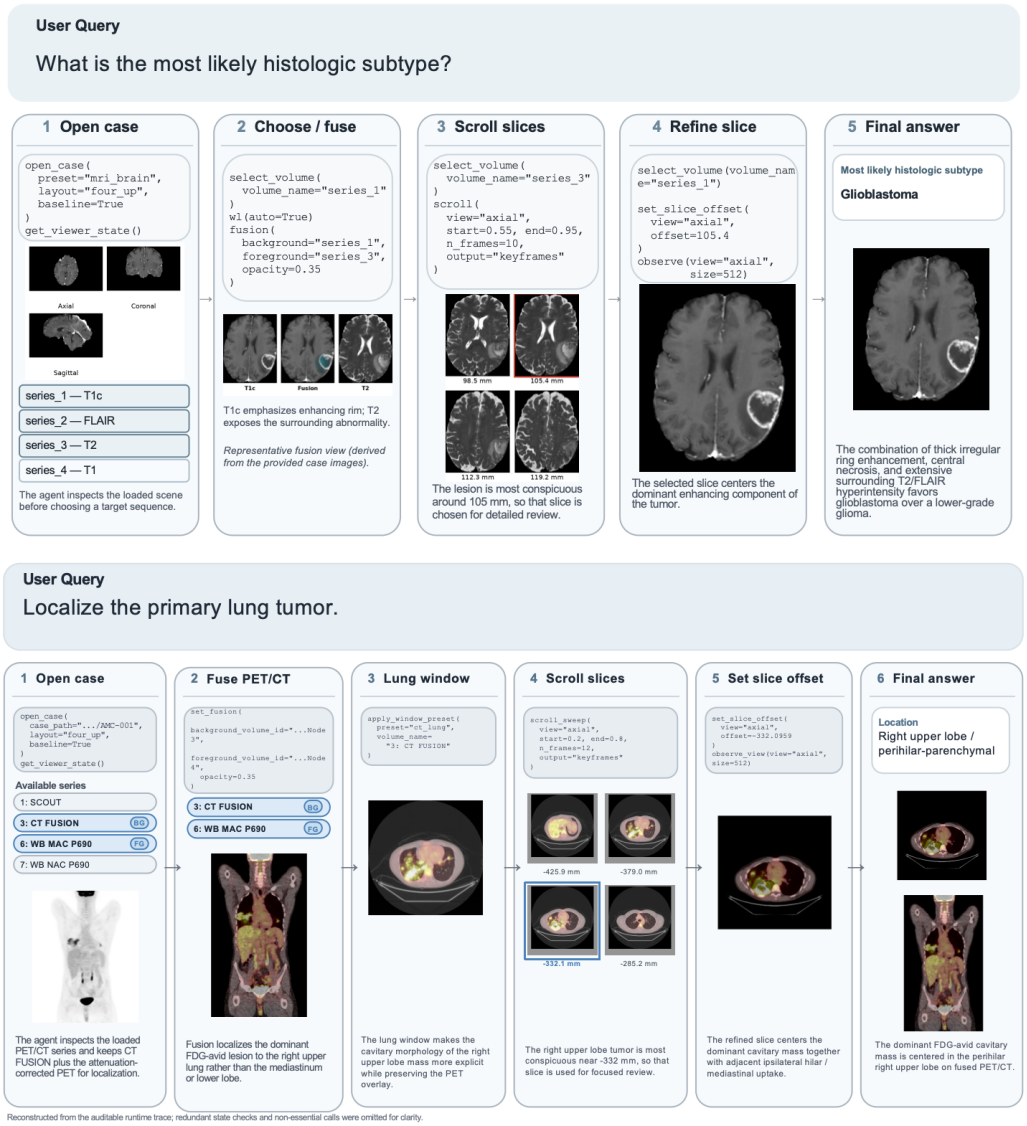

Figure 2: Representative auditable execution traces from the Brain MRI (**top**) and Lung CT/PET (**bottom**) modules under the Tool-Use setting. This demonstration compresses the actual longer runtime logs into briefer decision-relevant steps and workflows. The complete action chain, including the corresponding tool arguments, visual outputs, and final reports/answers are all auditable.

MEDFLOW-BENCH relies on this exact separation to distinguish pure viewer-native study inspection from complex, tool-augmented execution.

To maintain a bounded and legible interface, the callable surface remains strictly explicit. Functions already supported by 3D Slicer are wrapped via documented WebServer REST endpoints, which process HTTP requests and responses to allow external control. Operations that are not cleanly covered by this REST interface, such as DICOM import, quantitative measurement, and DICOM SEG export, are exposed through named bridge handlers. Crucially, this runtime is deliberately restrictive. While 3D Slicer includes an embedded Python console, allowing an agent to generate and execute arbitrary code would enlarge the attack surface, weaken auditability, and complicate deployment. Therefore, MEDOPENCLAW exclusively exposes predefined operations and prohibits the execution of raw Python scripts.

This bounded design is what inherently guarantees auditability. The runtime logs every tool invocation alongside its arguments, the resulting viewer-state snapshot, and any generated artifacts. These records

Benchmark / resource	Input contract	Multi-imaging modality benchmark	Cross-modality in case	Active exploration / request	Differential diagnosis	Agentic execution required
Conventional fixed-input imaging QA / report resources						
VQA-RAD [1], VQA-Med [46], SLAKE [2], PubMedVision / HuatuoGPT-Vision [3], OmniMedVQA [14], MedXpertQA-MM [4], PMC-VQA [15]	Static input	✓	×	×	×	×
Richer imaging benchmarks: stronger reasoning supervision, pre-selected slices/views, or full-study static input						
DrVD-Bench [47]	Static input	✓	×	×	×	×
NOVA [21]	Pre-selected static input	×	×	×	×	×
MedThinkVQA [18]	Pre-selected static input	✓	✓	×	✓	×
ReXGroundingCT [34]	Full-study static input	×	×	×	×	×
3D-RAD [48]	Full-study static input	×	×	×	×	×
ViMed-PET [49]	Full-study static input	✓	✓	×	×	×
Agent-leaning or tool-oriented imaging resources						
ChestAgentBench (MedRAX) [20]	Static input	×	×	×	×	×
REFUGE2 (used by MedAgent-Pro) [50, 33]	Static input	×	×	×	×	×
RadABench-Data / EvalPlat [37]	Synthetic dynamic tool env.	✓	×	×	×	✓
Nearest text-only comparators for sequential diagnosis						
MedCaseReasoning (text-only) [27]	Static text-only case	×	×	×	×	×
APP (text-only) [29]	Dynamic text-only case	×	×	✓	×	✓
Sequential Diagnosis Benchmark / MAI-DxO (text-only) [51]	Dynamic text-only case	×	×	✓	✓	✓
MEDFLOW-BENCH (ours)	Full-study interactive access	✓	✓	✓	✓	✓

Table 1: **Comparison of MEDFLOW-BENCH with prior imaging and text-only diagnosis benchmarks.** Multi-modality refers to imaging modalities, and active exploration denotes sequential evidence acquisition or study navigation. MEDFLOW-BENCH is the only benchmark here with full-study interactive access, cross-modality cases, active exploration, differential diagnosis, and agentic execution.

ensure the diagnostic trajectory is fully reconstructable after the fact, detailing which views were accessed, what actions were executed, and the specific evidence that supported the final answer.

As a concrete example, Figure 2 illustrates abbreviated, decision-relevant execution traces from the Brain MRI and Chest CT/PET modules. In the Brain MRI scenario, the agent sequentially enumerates available volumes, inspects major MRI sequences (T1c, FLAIR, T2, and T1), requests cross-sequence observations, and scrolls through axial keyframes before committing to a diagnosis. Across both examples, the query, tool calls, visual outputs, and response remain externally inspectable instead of collapsing into an opaque intermediate state.

Crucially, this observable, step-by-step execution does more than just benchmark model capabilities. It directly mirrors the workflow of a human radiologist. Because the interactions are transparent and bounded, the framework readily translates from an autonomous evaluation runtime into an interactive, human-in-the-loop assistant. This is the foundation of MEDCOPILOT, a clinician-facing application built directly on top of MEDOPENCLAW. By leveraging these exact auditable tool-use traces, MEDCOPILOT can autonomously handle cumbersome viewer interactions, such as fusing modalities or localizing key slices, thereby reducing manual overhead, improving workflow efficiency, and allowing clinicians to focus on final diagnostic verification.

4 MEDFLOW-BENCH: Study-Level Evaluation

Because existing static-image benchmarks cannot evaluate the dynamic, multi-step reasoning clinical workflow enabled by MEDOPENCLAW, we introduce MEDFLOW-BENCH to evaluate the full interactive loop. MEDFLOW-BENCH evaluates study-level reasoning rather than static-image perception. As highlighted in Table 1, it distinguishes itself from prior resources by offering full-study interactive access, cross-modality reasoning, and required agentic execution. The current initial release contains

two representative modules built from public datasets and scored under a shared episode definition. A benchmark episode is defined at the study level rather than the image level. Each episode specifies (i) a study package containing the full volumetric exam and study metadata, (ii) a task prompt that asks for a case-level or study-level decision, (iii) an allowed action space determined by the evaluation track, and (iv) a canonical answer schema used for scoring.

4.1 Modules and Evaluation Protocols

The current release spans two clinical modules evaluated under strict answering protocols. The brain MRI module uses UCSF-PDGM [52], a preoperative multi-sequence brain tumor MRI dataset, for case-level diagnosis over a fixed label set. The lung CT/PET module uses the NSCLC radiogenomics dataset [53], a paired non-small cell lung cancer CT/PET cohort with pathology annotations, for five structured prediction tasks: tumor location, pathological T stage, pathological N stage, histology, and histopathological grade. Each episode is evaluated under two answer protocols. One track is based on a Multiple-Choice question format (MCQ) which provides explicit options. The other track relies on an Open-ended question format which keeps the task unchanged but removes options, using an LLM judge against a canonical answer as a secondary measure of robustness. For brain MRI, we report case-level accuracy. For lung CT/PET, we report case-exact accuracy as the primary metric, with question-level accuracy as an auxiliary measure.

4.2 The Three-Track Design

To support both model evaluation and systems work, MEDFLOW-BENCH separates the solution space into three tracks rather than mixing all methods on one leaderboard. All tracks use identical cases, task formulations, and metrics.

- **Track A: Viewer-Only.** A test of pure full-study visual perception. Aligning with the first layer of our runtime architecture, methods use MEDOPENCLAW to drive 3D Slicer using only primitive tools (e.g., series selection, scrolling, windowing). By excluding expert tools, this track focuses on visual search, slice-to-slice synthesis, and sequence-level reasoning rather than tool engineering.
- **Track B: Tool-Use.** The main systems track, allowing unrestricted access to expert modules and evidence tools via MEDOPENCLAW. This track opens up the advanced layers of the runtime, testing whether a model can decide when an expert tool (like the MONAI pack) is needed, set parameters, and integrate the returned artifacts back into the diagnostic trajectory.
- **Track C: Open-Method.** Methods may bypass MEDOPENCLAW entirely and use any alternative pipeline that consumes the raw cases and outputs the canonical answer schema. We include this track to ensure the benchmark remains a universal standard rather than just a test of our specific runtime. This leaves room for future full-study paradigms such as native 3D foundation models, study compression encoders, or non-Slicer pipelines.

5 Experiments

To establish initial baselines for MEDFLOW-BENCH, we evaluate a suite of state-of-the-art vision-language models, including GPT-5.4, GPT-5-mini, Gemini-3-flash, and Gemini-3.1-pro. We report base models under identical prompts and track-specific tool budgets. The Multiple-Choice Question (MCQ) format serves as the evaluation protocol throughout for this initial version. For both Track A and Track B, all interactions go seamlessly through the MEDOPENCLAW runtime.

Track A: Viewer-Native Baselines. Table 2 presents a controlled comparison of base models operating under identical viewer-native access, restricting them to purely primitive tools like scrolling and windowing. The results demonstrate that frontier models can already navigate the viewer to solve a meaningful portion of study-level tasks. For example, in the Brain MRI module, Gemini-3.1-pro achieves the highest case-level accuracy at 0.63, closely followed by GPT-5.4 at 0.61. However, the detailed sub-metrics for the Lung CT/PET module reveal that while models perform moderately well at macroscopic tasks like Tumor Location (where Gemini-3.1-pro achieves 0.43 and GPT-5.4 achieves 0.46), performance drops precipitously on complex, fine-grained tasks. Histopathological Grade prediction, for instance, remains exceptionally challenging, with all tested models struggling to surpass baseline random chance.

Model	Brain MRI	Lung CT/PET (Chest)					
		Overall	Tumor Location	T Stage	Pathological N Stage	Histology	Histopath. Grade
MCQ Protocol							
GPT-5.4	0.61 (5.9)	0.32 (11.5)	**0.46**	**0.32**	0.38	0.36	0.07
GPT-5-mini	0.43 (2.24)	0.20 (1.85)	0.14	0.17	0.09	0.06	0.04
gemini-3.1-flash-preview	0.56 (9.6)	**0.52** (19.6)	0.42	0.21	**0.83**	**0.72**	**0.44**
gemini-3.1-pro-preview	**0.63** (7.2)	0.31 (11.7)	0.43	0.31	0.35	0.34	0.11

Table 2: **Viewer-Only track.** Controlled comparison of base models under identical viewer-native access. Lung CT/PET (Chest) evaluates overall performance alongside five sub-metrics. We show Accuracy (Avg. Tool Calls) for Brain MRI and overall Lung CT/PET, and Accuracy for the Lung CT/PET sub-metrics. The highest scores are marked in bold.

Backbone	System	Brain MRI (MCQ)	Lung CT/PET (MCQ)
GPT-5-mini	MedOpenClaw with primitive tools	0.43	0.20
	+ With segmentation toolpacks	0.45	0.14
GPT-5.4	MedOpenClaw with primitive tools	**0.61**	**0.32**
	+ With segmentation toolpacks	0.57	0.27

Table 3: **Tool-Use track.** Reference-system ablation study shows the performance drop-off by progressively adding segmentation toolpacks. The highest scores are marked in bold.

Track B: The Tool-Use Bottleneck. Moving from primitive viewer actions to advanced expert analysis, our preliminary results reveal a counterintuitive trend: equipping the agent with the Segmentation Toolpack does not currently improve overall accuracy, and in some cases, even degrades it. As shown in Table 3, when provided with segmentation toolpacks, GPT-5.4's accuracy on the Brain MRI module drops from 0.61 to 0.57, and its performance on the Lung CT/PET module falls from 0.32 to 0.27.

This suggests that while the underlying segmentation algorithms are robust, current VLM agents have not yet developed the precise spatial grounding required to operate them effectively. For example, when invoking the Local Threshold Segmentation Tool, the agent must provide accurate spatial coordinates to guide the algorithm. Because current models often struggle to output these spatial coordinates with millimeter-level precision, the tool frequently generates misaligned or anatomically incorrect masks. Consequently, the agent ends up relying on flawed, self-generated visual evidence, which misleads its subsequent diagnostic reasoning. This finding highlights a critical bottleneck in the Tool-Use setting: providing reliable expert tools is insufficient if the agent's foundational control capabilities are still maturing.

6 Discussion and Conclusion

Discussion. The introduction of MEDOPENCLAW and MEDFLOW-BENCH shifts the evaluation paradigm of medical vision-language models from isolated, static-image recognition to dynamic, full-study reasoning. Our initial findings highlight several critical implications for the future of medical AI.

- **The "Tool-Use Paradox" and Spatial Grounding:** Perhaps our most significant empirical finding is that providing current VLMs with advanced, expert-level segmentation tools often degrades their overall diagnostic accuracy. While the AI community has made massive strides in logical and language-conditioned reasoning, our benchmark exposes a critical gap in fine-grained spatial grounding. Reliable tool-augmented execution remains highly challenging because models cannot yet output the precise, millimeter-level coordinates required to seed clinical algorithms. Solving this spatial control bottleneck is the next great frontier for medical agents.
- **Auditability as a Prerequisite for Clinical Trust:** For AI to be deployed in real-world hospitals, black-box decision-making is unacceptable. By restricting the VLM to a bounded runtime where

every series selection, window adjustment, and slice scroll is explicitly logged, MEDOPENCLAW naturally produces the transparent evidence trail that clinical and regulatory frameworks demand.

- **Bridging Benchmarks and Real-World Applications:** Unlike static 2D benchmarks where high scores often fail to translate to clinical utility, solving MEDFLOW-BENCH directly contributes to building better human-in-the-loop systems. Because the action space perfectly mirrors the workflow of a human radiologist, improvements on this benchmark directly enhance the capabilities of downstream clinical assistants like MEDCOPILOT.

Limitations and Roadmap. We consider this current iteration to be a foundational **first release**. By establishing the runtime infrastructure and defining the study-level episode protocol across two representative clinical modules, we lay the groundwork for a broader, community-driven ecosystem. Future releases will aggressively expand this foundation by 1) scaling modalities: expanding the case diversity beyond multi-sequence Brain MRI and Lung CT/PET to include ultrasound, mammography, and longitudinal studies (comparing prior and current exams). 2) Broadening evaluation settings: introducing multi-turn conversational evaluation tracks and tasks requiring the synthesis of extensive patient electronic health records (EHR) alongside the imaging. 3) Enriching the tool ecosystem: integrating a wider array of specialized algorithms from the MONAI ecosystem to further test model capabilities as spatial grounding improves.

Conclusion. Medical agents must do more than answer localized questions about pre-selected, curated images. They must actively navigate full 3D studies, gather comprehensive information, and leave an auditable trail of evidence for their decisions. MEDOPENCLAW provides a secure, bounded runtime for this complex interaction, and MEDFLOW-BENCH offers a rigorous, controlled environment to evaluate it. Our current results suggest that viewer-native full-study reasoning is already a tangible reality for frontier VLM agents, even as reliable, tool-augmented execution remains a compelling challenge for the community to solve. .

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